

PROJECT REPORT – 1

OMNICHANNEL RETAIL SALES & INVENTORY DASHBOARD

1. Introduction

The Omnichannel Retail Sales & Inventory Dashboard is a business intelligence project developed to analyze retail sales performance, customer purchasing trends, inventory management, and product availability across different cities and sales channels.

The main purpose of this project is to help retail businesses monitor their sales activities in real time and make better business decisions using data-driven insights. The dashboard combines sales data from multiple channels such as Online, App, and Store sales into a single interactive report.

This project was developed using:

- Power BI for dashboard creation and data visualization
- SQL for data extraction and querying
- Python for data cleaning and Exploratory Data Analysis (EDA)

The dashboard allows users to filter and analyze data based on date range, city, product category, and sales channel.

2. Problem Statement

Retail businesses generate large amounts of sales and inventory data every day. Managing this data manually becomes difficult and time-consuming.

The organization needed a solution to:

- Track total sales and orders
- Monitor inventory availability
- Identify top-performing products
- Compare sales across different cities
- Analyze customer purchasing channels
- Improve stock management
- Support business decision-making with real-time insights

Without proper visualization and analysis, businesses may face problems such as overstocking, stock shortages, poor sales performance, and ineffective inventory planning.

3. Objectives of the Project

The major objectives of this project are:

1. To analyze omnichannel retail sales performance
 2. To monitor inventory levels across products
 3. To identify high-performing products and cities
 4. To compare sales across Online, App, and Store channels
 5. To provide interactive business insights using dashboards
 6. To support decision-making using real-time data visualization
 7. To improve inventory optimization and sales performance
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4. Technologies Used

Power BI

Power BI was used to:

- Create interactive dashboards
- Build charts and KPI cards
- Add filters and slicers
- Generate business insights
- Design data visualizations

SQL

SQL was used for:

- Extracting retail data
- Filtering and querying records
- Managing structured datasets
- Performing data aggregation

Python

Python was used for:

- Data cleaning
- Handling missing values

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Identifying trends and patterns

Libraries commonly used:

- Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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5. Dataset Description

The dataset contains retail sales and inventory information collected from different cities and sales channels.

The important fields included:

- Order ID
- Product Name
- Product Category
- Sales Amount
- Quantity Sold
- City
- Sales Channel
- Stock Available
- Order Date
- Customer Details

The dataset helps in analyzing sales performance, inventory status, and customer buying behavior.

6. Data Cleaning and Preprocessing

Before creating the dashboard, the dataset was cleaned and prepared using Python and SQL.

The following preprocessing steps were performed:

- Removed duplicate records
- Handled missing values
- Corrected inconsistent data entries
- Converted date columns into proper format
- Standardized category names
- Verified numerical values

- Filtered invalid records

This process improved data quality and ensured accurate dashboard results.

7. Exploratory Data Analysis (EDA)

EDA was performed to understand sales patterns, customer behavior, and inventory trends.

The following analyses were carried out:

Sales Trend Analysis

Sales performance was analyzed year-wise to identify growth patterns and business performance over time.

Channel-wise Analysis

Sales were compared across:

- Online
- App
- Store

This helped identify the most profitable sales channel.

Product Performance Analysis

Products with higher sales and quantity sold were identified.

Inventory Analysis

Stock availability was analyzed to understand inventory levels and avoid stock shortages.

City-wise Sales Analysis

Sales performance across different cities was compared to identify top-performing regions.

Category-wise Analysis

Different product categories such as:

- Clothing
- Electronics
- Sports

- Home

were analyzed based on total sales contribution.

8. Dashboard Overview

The dashboard contains multiple sections and visualizations that provide complete business insights.

KPI Cards

The dashboard displays important KPIs such as:

- Total Sales: 31.99M
- Total Orders: 5044
- Total Quantity: 13K
- Average Order Value (AOV): 6.34K

These KPIs provide a quick overview of overall business performance.

Sales by Year

A line chart was used to analyze sales trends from 2022 to 2024.

Insights:

- Sales increased significantly in 2023
- Slight decrease observed in 2024
- Overall business performance remained strong

Sales by Channel

A donut chart was used to compare sales distribution across:

- Online
- App
- Store

Insights:

- Online channel contributed the highest sales
- App and Store sales also contributed significantly
- Omnichannel strategy improved overall business reach

Stock Left Based on Product

Bar charts were used to analyze remaining inventory for products such as:

- Shirt
- Phone
- Headphones
- Shoes
- Watch
- Bag
- Laptop
- Sofa

Insights:

- Inventory levels were balanced across products
- Some products required better stock optimization

Total Sales by Category

A treemap visualization was used to compare category-wise sales.

Insights:

- Clothing and Electronics contributed major revenue
- Sports and Home categories also showed stable performance

Total Sales by City

City-wise sales comparison helped identify high-performing regions.

Insights:

- Mumbai and Hyderabad generated strong sales
- Regional analysis helps businesses target profitable markets

Product Summary Table

A detailed table displayed:

- Product name
- Total sales
- Quantity sold
- Available stock

This helped in identifying best-selling products and stock status.

9. Key Insights

The following important insights were identified:

1. Online sales channel generated the highest revenue.
2. Clothing and Electronics categories contributed major sales.
3. Some products showed high sales with low remaining inventory, indicating the need for stock replenishment.
4. Mumbai and Hyderabad performed well in terms of total sales.
5. Sales growth was highest in 2023.
6. Inventory tracking helped identify stock availability across products.
7. Omnichannel sales strategy improved customer reach and revenue generation.

10. Business Benefits

This dashboard provides several business benefits:

- Real-time sales monitoring
- Better inventory management
- Improved decision-making
- Identification of top-performing products
- Better regional sales analysis
- Enhanced customer purchasing analysis
- Faster business reporting
- Data-driven strategic planning

11. Challenges Faced

Some challenges faced during the project were:

- Cleaning inconsistent data
- Handling missing values
- Managing large datasets
- Designing interactive visuals

- Creating proper relationships between tables
- Optimizing dashboard performance

These challenges were solved using Python preprocessing, SQL querying, and Power BI optimization techniques.

12. Final Conclusion

The Omnichannel Retail Sales & Inventory Dashboard successfully provides a complete overview of retail sales and inventory performance.

The project helps businesses:

- Monitor sales activities in real time
- Analyze customer purchasing behavior
- Track inventory availability
- Compare sales across cities and channels
- Identify high-performing products
- Improve operational efficiency

The integration of Power BI, SQL, and Python made the project highly interactive, analytical, and business-oriented.

Overall, this dashboard can help retail companies improve profitability, inventory planning, and business decision-making.

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PROJECT REPORT – 2

HOTEL BOOKING ANALYTICS DASHBOARD

1. Introduction

The Hotel Booking Analytics Dashboard is a business intelligence project designed to analyze hotel booking trends, customer behavior, cancellation patterns, seasonal demand, and revenue performance.

The project helps hotel management understand customer booking patterns and improve operational efficiency through data-driven insights.

The dashboard provides a detailed overview of:

- Total bookings
- Confirmed bookings
- Cancellation rate
- Average ADR (Average Daily Rate)
- Customer segmentation
- Seasonal demand
- Booking channels
- Market segment analysis

The project was developed using:

- Power BI
 - SQL
 - Python
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2. Problem Statement

Hotels receive thousands of bookings from different customer types and booking platforms. Managing and analyzing this information manually becomes difficult.

The organization required a dashboard to:

- Monitor booking trends
- Track cancellation rates
- Understand customer behavior
- Analyze seasonal demand
- Improve revenue management
- Identify profitable customer segments
- Optimize hotel operations

Without proper analysis, hotels may face issues such as:

- High cancellation rates
 - Poor room utilization
 - Revenue loss
 - Inefficient pricing strategies
 - Improper demand forecasting
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3. Objectives of the Project

The main objectives of this project are:

1. To analyze hotel booking performance
 2. To monitor cancellation trends
 3. To study seasonal customer demand
 4. To identify profitable customer segments
 5. To improve pricing and revenue strategies
 6. To provide interactive business insights using Power BI
 7. To support hotel decision-making using analytics
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4. Technologies Used

Power BI

Used for:

- Dashboard design
- Data visualization
- KPI creation
- Interactive filtering
- Report generation

SQL

Used for:

- Data extraction
- Query execution
- Aggregation and filtering
- Managing hotel booking datasets

Python

Used for:

- Data cleaning
- Exploratory Data Analysis
- Data preprocessing
- Trend identification

Libraries used:

- Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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5. Dataset Description

The hotel booking dataset contains information related to hotel reservations and customer details.

The important fields included:

- Booking ID
- Customer Type
- Reservation Status
- Market Segment
- ADR (Average Daily Rate)
- Country
- Season
- Booking Channel
- Cancellation Status
- Room Type
- Booking Date

This dataset helps in analyzing customer behavior and hotel business performance.

6. Data Cleaning and Preprocessing

Before visualization, the data was cleaned using Python and SQL.

The preprocessing steps included:

- Removing duplicate records
- Handling missing values
- Standardizing category names
- Converting date formats
- Correcting inconsistent entries
- Filtering invalid booking records

Proper preprocessing improved dashboard accuracy and reliability.

7. Exploratory Data Analysis (EDA)

EDA was conducted to understand booking behavior and hotel demand patterns.

The analyses included:

Booking Trend Analysis

Total bookings and confirmed bookings were analyzed to understand hotel demand.

Cancellation Analysis

Cancellation patterns were studied to identify factors affecting booking cancellations.

Seasonal Demand Analysis

Booking trends were analyzed across:

- Autumn
- Monsoon
- Summer
- Winter

This helped identify peak and low-demand seasons.

ADR Analysis

Average Daily Rate (ADR) was analyzed across different seasons and customer types.

Market Segment Analysis

Bookings were analyzed based on market segments such as:

- Online TA
- Offline TA
- Corporate
- Direct
- Groups

Customer Type Analysis

Customer categories were compared to understand booking behavior and revenue contribution.

8. Dashboard Overview

The dashboard contains multiple KPI cards and visualizations.

KPI Cards

The dashboard displays:

- Total Bookings: 87K
- Confirmed Bookings: 63K
- Cancellation Rate: 0.27
- Average ADR: 106.27

These KPIs provide a quick overview of hotel performance.

Average ADR by Season

A bar chart was used to compare ADR across different seasons.

Insights:

- Some seasons generated higher ADR values
- Seasonal pricing trends were identified
- Demand fluctuations affected room pricing

ADR by Cancellation Status

This visualization compared ADR between canceled and confirmed bookings.

Insights:

- Cancellation patterns impacted hotel revenue
- High cancellation rates affected occupancy planning

Stay Length vs ADR

A scatter plot was used to analyze the relationship between stay length and ADR.

Insights:

- Customers with longer stays contributed higher revenue in some cases
- Pricing trends varied across customer groups

Booking by Customer Type

A donut chart was used to compare bookings by customer type.

Insights:

- Some customer categories contributed higher booking percentages
- Customer segmentation helped understand hotel demand

Total Bookings by Market Segment

A treemap visualization analyzed booking distribution across market segments.

Insights:

- Online travel agencies generated major bookings
- Corporate and direct bookings also contributed significantly

Seasonal Customer Table

A detailed table displayed ADR values across seasons and customer categories.

This helped compare customer spending patterns during different seasons.

9. Key Insights

The following insights were identified:

1. Online travel agencies generated a large share of bookings.
 2. Seasonal demand strongly affected hotel bookings and ADR.
 3. Cancellation rates impacted occupancy and revenue planning.
 4. Different customer types showed different spending patterns.
 5. Higher ADR values were observed during peak demand seasons.
 6. Customer segmentation improved understanding of booking behavior.
 7. Booking analysis helped improve operational planning.
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10. Business Benefits

The dashboard provides several advantages:

- Better booking analysis
 - Improved revenue management
 - Seasonal demand forecasting
 - Better customer understanding
 - Improved pricing strategies
 - Occupancy optimization
 - Faster business reporting
 - Data-driven hotel management decisions
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11. Challenges Faced

Some challenges during the project included:

- Cleaning inconsistent booking data
- Managing missing values
- Handling large datasets
- Designing interactive reports
- Analyzing cancellation trends
- Creating efficient Power BI relationships

These issues were resolved using Python preprocessing and Power BI optimization.

12. Final Conclusion

The Hotel Booking Analytics Dashboard successfully provides valuable insights into hotel booking performance, customer behavior, cancellation trends, and seasonal demand.

The project helps hotel management:

- Monitor booking trends
- Improve revenue management
- Optimize pricing strategies
- Understand customer behavior
- Reduce cancellation impact
- Improve occupancy planning
- Support better decision-making

The combination of Power BI, SQL, and Python made the project interactive, analytical, and business-focused.

Overall, this dashboard can help hotels improve operational efficiency, customer satisfaction, and overall profitability.